
Compact-gauge Bloch oscillations and classical thermalization bound real-time lattice scalar electrodynamics of strong-field plasmas

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Abstract

Real-time lattice scalar quantum electrodynamics (QED) advances an ensemble of classical Klein–Gordon–Maxwell fields to model plasmas in which relativistic-quantum scales overlap collective scales. We build a variational Yee-type solver whose discrete $U(1)$ invariance conserves charge to machine precision ($\sim 10^{-14}$) and whose exterior-calculus construction makes the Bianchi identities exact ($\sim 10^{-13}$), and we validate it against the two published demonstrations: the unmagnetized dispersion relations and the transition from laser wakefield acceleration to Schwinger pair production, recovering the reported few-parts-per-million and sub-percent energy budgets. We then map both the reach and the intrinsic limits of the method. On the reach side, the covariant Wilson-line coupling reproduces the relativistic dressed mass $m_* = m\sqrt{1 + a_0^2}$ of a boson in an intense circularly polarized field to better than 0.1% with a discretization error that falls as the square of the transverse cell size—quantitatively predicting the relativistic self-induced transparency cutoff—while a self-neutral electron–positron pair plasma reproduces its collective mode $\omega^2 = k^2 + 2\omega_{p1}^2$ with $\mathcal{O}(\Delta x^2)$ -convergent error and machine-precision per-species charge conservation. On the limit side we identify two obstructions that are framework-level rather than numerical, demonstrate that each persists across resolutions, and repair each at the level of the action. First, the compactness that protects charge confines charged modes to a finite Brillouin zone, so a sustained accelerating field drives a spurious Bloch oscillation—the current reverses at $t_{\text{rev}} = \pi/(qE_0\Delta x)$ instead of saturating; a non-compact covariant derivative removes the reversal at the cost of relaxing local charge conservation to $\mathcal{O}(\Delta x^2)$. Second, the classical-statistical sampling that makes the scheme tractable thermalizes to Rayleigh–Jeans equipartition rather than Bose–Einstein, so the spectral centroid tracks the grid cutoff ($\langle\omega\rangle \propto 1/\Delta x$) and has no continuum limit; a quantum fluctuation–dissipation thermostat restores a Bose–Einstein spectrum and a finite continuum limit, trading exact energy conservation for detailed balance while preserving charge and gauge invariance.

1 Introduction

When a plasma becomes dense enough, or a field strong enough, that its collective length and time scales approach the Compton wavelength and time, the kinetic description that underpins conventional plasma physics—itsself a geometrical-optics limit of an underlying relativistic-quantum theory [Ruiz and Dodin, 2015]—ceases to be self-consistent. Intense-laser pair production [Liang et al., 1998, Sarri et al., 2015, Di Piazza et al., 2012, Fedotov et al., 2023], wave dynamics on Compton scales, and astrophysical plasmas in critical magnetic fields all lie in this regime. Semiclassical re-

pairs insert prefabricated source terms into fluid or kinetic equations [Kluger et al., 1998, Schmidt et al., 1998, Hebenstreit et al., 2010], but they cannot capture the interplay of coexisting processes or their quantum coherence, and they offer no consistent way to remove both energy and momentum from the fields once many pairs are made [Schützhold et al., 2008]. A faithful account must live at the relativistic-quantum level. Full path-integral lattice QED is prohibitive, but in the classical-statistical regime of high occupation and weak coupling ($e \approx 0.3$) tree-level physics dominates and the quantum evolution is well approximated by advancing classical field equations over an ensemble of statistically equivalent initial conditions [Aarts and Berges, 2002, Mueller and Son, 2004, Jeon, 2005, Berges et al., 2014, Gelis and Tanji, 2013, Gelfand et al., 2016].

Building on this idea, Shi et al. [2018] developed a real-time lattice *scalar* QED that sidesteps fermion doubling and isolates unambiguous plasma contributions: a Yee-type discrete action couples a scalar field on vertices to a gauge one-form on edges, and its Euler–Lagrange equations are a discrete Klein–Gordon equation, a discrete Gauss law, and a discrete Maxwell–Ampère law. The construction has two structural virtues. Its exact discrete U(1) invariance makes local charge conservation hold to machine precision, and its exterior-calculus discretization makes the Bianchi identities exact by construction. The method was demonstrated on two problems—linear-wave spectra and a gamma-ray laser striking a dense slab—that together cover only a narrow slice of its claimed scope.

This paper asks two questions that the original demonstrations leave open. *How far does the method reach?* We show that its covariant coupling faithfully captures two effects it was never tested on: the relativistic dressed mass that a boson acquires in an intense field, and the collective modes of a self-neutral pair plasma—the very state that the method’s own pair-production channel produces. *Where, and why, does it fail?* We identify two obstructions that are not bugs and not under-resolution but consequences of the framework’s defining choices, and we repair each at the level of the action. The first is topological: the compact gauge group that protects charge also compactifies momentum space into a finite Brillouin zone, so a sustained accelerating field produces Bloch oscillations rather than acceleration. The second is statistical: the classical-statistical sampling that makes the method tractable relaxes to classical equipartition, so the late-time spectrum is Rayleigh–Jeans, not Bose–Einstein, and worsens as the grid is refined. Both obstructions are, we argue, generic to real-time lattice field theory for plasmas; understanding them, and knowing how to relax them, is a prerequisite for trusting the method’s long-time and strong-field predictions.

2 Methodology

2.1 Variational lattice scalar QED and our implementation

We work in natural units $\hbar = c = \epsilon_0 = 1$ and set the boson mass $m = 1$. In the continuum the model is the Klein–Gordon–Maxwell system for a complex scalar ϕ of charge q coupled to a gauge field A_μ through the covariant derivative $D_\mu = \partial_\mu - iqA_\mu$, derived by extremizing $\mathcal{L} = (\overline{D_\mu \phi})(D^\mu \phi) - m^2 \phi \phi - \frac{1}{4} F_{\mu\nu} F^{\mu\nu}$. Following Shi et al. [2018] we discretize on a rectangular lattice with ϕ_v on vertices and the gauge one-form A_e on edges. Covariant derivatives use Wilson lines, e.g. $(D_1^<\phi)_{i+1/2}^n = \Delta x^{-1}(\phi_{i+1}^n e^{-iq\Delta x A_{i+1/2}^n} - \phi_i^n)$, and the field strength is the exact lattice curvature $F = dA$ on faces. Extremizing the discrete action $S_d = \sum_c \Delta V L_d[\phi_v, A_e]$ yields the discrete Klein–Gordon equation [Eq. (16) of Shi et al., 2018], the discrete Gauss law [Eq. (17)], and the discrete Maxwell–Ampère law [Eq. (19)]. We implement the explicit temporal-gauge scheme (with a Lorenz-gauge option) exactly as in their Appendix B; the charged field is advanced by

$$\phi_i^{n+1} = 2\phi_i^n - \phi_i^{n-1} + \Delta t^2 \left[\frac{\phi_{i+1}^n e^{-iq\Delta x A_{i+1/2}^n} - 2\phi_i^n + \phi_{i-1}^n e^{+iq\Delta x A_{i-1/2}^n}}{\Delta x^2} - (m^2 + M_{\perp,i}^2) \phi_i^n \right], \quad (1)$$

where the transverse Wilson-line hops contribute the lattice dressed mass $M_{\perp,i}^2 = \sum_{\ell=y,z} (4/\Delta_\perp^2) \sin^2(q\Delta_\perp A_i^\ell/2)$, and the spatial links by the discrete Ampère law $A_{s+\ell/2}^{n+1} = 2A_{s+\ell/2}^n - A_{s+\ell/2}^{n-1} + \Delta t^2(\partial^2 A + J_{s+\ell/2}^n)$ with the covariant currents of Eq. (20). Charged species are transverse-uniform, so the y, z hops use $\phi_{s\pm\pm} = \phi_s$ and reduce to one value per vertex; $\Delta_\perp$ is the transverse cell size, which enters only the *nonlinear* coupling. Our solver, its validation, and every run script are provided in `proposal/code`; no external lattice-field-theory code is used.

2.2 Implementation verification

Before any physics run we verified the four structural properties on which every downstream claim rests (`logs/validation.json`): (i) total charge is conserved to 1.9×10^{-14} under a random gauge drive; (ii) the discrete Gauss law is preserved to 5.4×10^{-10} (relative to the initial residual, i.e. machine precision accumulated over 3000 steps); (iii) the discrete Faraday identity $\partial_t B = -(\nabla \times E)$ holds to 9.2×10^{-14} , confirming the Bianchi identities by construction; and (iv) the total energy is conserved *exactly* in the free limit (5.5×10^{-16} at $q = 0$) and only to $\mathcal{O}(q\Delta t^2)$ at finite coupling, exactly as Shi et al. [2018] predict from the Yee-type action. The free and cold-plasma dispersion relations are recovered to 3.1×10^{-4} and 2.6×10^{-5} respectively. This asymmetry—charge exact, energy only second order—is not incidental; it recurs as the organizing theme of our results.

2.3 Predictions, failure criteria, diagnostics, and tolerances

We study four settings beyond the published demonstrations. Each pairs the simulation with a sharp prediction, stated here before any result, and each is designed so that a distinct class of discretization error is its *primary diagnostic*: the covariant ϕ – A coupling, dispersion fidelity, gauge invariance, and energy conservation, respectively. Tolerances are fixed once, below, and used unchanged in Sec. 4.

(P1) Relativistic dressed mass and induced transparency (primary diagnostic: ϕ – A coupling).

A boson bathed in a circularly polarized field of normalized amplitude $a_0 = qA_0/m$ acquires the dressed mass $m_*^2 = m^2 + q^2 \langle A^2 \rangle$; for circular polarization $\langle A^2 \rangle = A_0^2$, giving the sharp prediction

$$m_*/m = \sqrt{1 + a_0^2}, \quad \text{hence the transparency cutoff } \omega_{co}/\omega_p = (1 + a_0^2)^{-1/4}, \quad (2)$$

since the cutoff obeys $\omega_{co}^2 = \omega_p^2/\gamma_0$ with $\gamma_0 = m_*/m$ [Eliasson and Shukla, 2011, Shi et al., 2016]. This is a pure Wilson-line effect: on the lattice the continuum $q^2 A^2$ is replaced by $(4/\Delta_\perp^2) \sin^2(q\Delta_\perp A/2)$, so the *tolerance* is the resulting cycle-averaged discretization error, which we predict scales as $\mathcal{O}(\Delta_\perp^2)$ and verify by a $\Delta_\perp$ scan. Agreement is claimed if the measured m_*/m matches Eq. (2) to within this error ($<1\%$ at the paper resolution $m\Delta_\perp = 0.04$).

(P2) Collective modes of a pair plasma (primary diagnostic: dispersion fidelity). An electron–positron pair plasma—equal densities n of charges $\pm q$, no neutralizing background—is self-neutral, and both species are dynamical, unlike the single mobile species over immobile ions of Shi et al. [2018]. Its plasma frequency is the incoherent sum, giving the transverse EM branch

$$\omega^2 = k^2 + \omega_{p,\text{pair}}^2, \quad \omega_{p,\text{pair}}^2 = 2q^2 n/m, \quad (3)$$

and electrostatic branches from $1 + \chi_+ + \chi_- = 0$ [Eq. (23) of Shi et al., 2018 summed over species]. The *tolerance* is the lattice dispersion error, which we predict scales as $\mathcal{O}(\Delta x^2)$ and verify by a Δx scan; agreement is claimed if the measured transverse branch matches Eq. (3) to within it ($<1\%$ at $m\Delta x = 0.04$).

(B1) Sustained-field Bloch oscillation (primary diagnostic: gauge invariance). The scheme’s gauge field lives on a torus: the theory is invariant under $A_e \rightarrow A_e + 2\pi/(q\Delta)$, so all observables are periodic in $qA\Delta$. Equivalently, charged modes live in a *finite Brillouin zone*. We drive a spatially uniform plasma with a sustained, uniform external field E_0 (temporal gauge: $A_x(t) = -E_0 t$). In the continuum the longitudinal current is monotonic, $j(t) = q^2 n E_0 t / \sqrt{(qE_0 t)^2 + m^2}$, and never reverses. Our *failure criterion*, stated in advance: the lattice current $J_x(t) = (2q/\Delta x) |\phi_0|^2 \sin(qE_0 \Delta x t)$ *reverses sign* at

$$t_{\text{rev}} = \pi/(qE_0 \Delta x), \quad (4)$$

a spurious deceleration absent from the continuum. The violated assumption is the compact (torus) topology, equivalently $qA\Delta < 2\pi$; the failure is fundamental because t_{rev} scales as $1/\Delta x$ and is present at every resolution (Sec. 4.4). The *tolerance* on t_{rev} is the frequency resolution of the current time series ($\pm$ one sample), and the failure is confirmed if t_{rev} matches Eq. (4) to within a few percent at all Δx .

(B2) Classical Rayleigh–Jeans thermalization (primary diagnostic: energy conservation). The classical-statistical approximation replaces field operators by c-numbers, dropping the commutator ($+\frac{1}{2}$ per mode) that is negligible only at high occupation. Its equilibrium is therefore classical equipartition (energy per mode $E_k \rightarrow k_B T$, i.e. Rayleigh–Jeans) rather than Bose–Einstein ($E_k = \omega_k / (e^{\omega_k/T} - 1)$) [Aarts and Berges, 2002, Berges, 2004]. Our *failure criterion*, stated in advance: the late-time energy-per-mode approaches a k -independent constant, so the energy-weighted spectral centroid tracks the grid cutoff,

$$\langle \omega \rangle \propto \pi / \Delta x, \quad (5)$$

i.e. it diverges as $\Delta x \rightarrow 0$ and the classical thermal state has *no continuum limit*. This is the sharp, *anti-convergent* signature that distinguishes a framework-level failure from a bug: a bug improves with resolution, this worsens. Energy is still conserved (it is the *distribution* that is wrong), so the diagnostic is the partition of the conserved energy. The failure is confirmed if the centroid-to-cutoff ratio $\langle \omega \rangle \Delta x / \pi$ is resolution-independent (equipartition) while the flatness of E_k is $\lesssim 0.3$.

For both boundary tests we (i) cross-check that the two published demonstrations succeed in the *same* code, ruling out an implementation artifact; (ii) run a convergence study at $m\Delta x \in \{0.08, 0.04, 0.02\}$ (B1) or $\{0.16, 0.08, 0.04\}$ (B2); (iii) reason from first principles; and (iv) propose, implement, and verify a fix documented at the level of the action, stating which invariants it preserves and which it relaxes. The fixes are derived in Secs. 4.4–4.5.

3 Experimental Setup

All runs use $q = 0.3$ (so $q^2/4\pi \approx 1/137$). **Linear waves (benchmark).** Single species, $\omega_p = 0.85m$ ($n_0 = 8.03 m^3$), immobile neutralizing ions, $m\Delta x = 0.04$, $m\Delta t = 0.02$, $L = 512$, $T = 1024$, ensemble 100; a global random carrier phase per realization and white-noise gauge fields ($\sigma = 10^{-4}m$) seed a broad spectrum, read out by a windowed 2D FFT. **Laser-plasma (benchmark).** $\omega_0 = 0.7m$, density plateau $n_0 = m^3$ ($\omega_p = 0.3m$) of width $100/m$ with Gaussian off-ramps $\sigma = 20/m$, linearly polarized pulse $\tau = 20/m$; $m\Delta x = 0.04$, $m\Delta t = 0.005$, box $280/m$, run to $tm = 300$, ensemble 48; temporal gauge, periodic boundaries. **(P1)** external circularly polarized drive of amplitude $a_0 \in \{0, 0.5, 1, 2, 4, 8\}$, boson oscillation frequency read from the phase slope; $\Delta_\perp$ scan $\{0.08, 0.04, 0.02, 0.01\}$; 6-member noise ensembles. **(P2)** two species $\pm q$, $n = 8m^3$ each ($\omega_{p,\text{pair}} = 1.2m$), otherwise as the linear-wave benchmark; Δx scan $\{0.08, 0.04, 0.02\}$. **(B1)** $E_0 = m^2/(3q)$ ($= 0.3E_c$, sub-Schwinger), $m\Delta x \in \{0.08, 0.04, 0.02\}$; the demonstration uses $m\Delta t = 0.005$. **(B2)** dense plasma with strong broadband transverse excitation ($qA \sim 1.5$), $m\Delta x \in \{0.16, 0.08, 0.04\}$, run to $tm \approx 500$; the thermostat fix uses a weakly nonlinear field ($qA \sim 0.05$) and bath temperature $T = 1.5m$. Sizes were chosen to keep peak memory well under the 50 GB budget; the embarrassingly parallel ensembles run across 32 cores.

4 Experimental Results

Table 1 collects the headline numbers; every value traces to a file in `proposal/logs` (collected by `collect_numbers.py` into `numbers.json`). We discuss each setting in turn, organized by physics.

4.1 Benchmarks: linear waves and laser-driven pair production

The ensemble-averaged power spectra of E_y and E_x (Fig. 1) reproduce Fig. 4 of Shi et al. [2018]. The transverse spectrum traces $\omega^2 = \omega_p^2 + k^2$ from the cutoff $\omega_p = 0.85m$ onto the light cone; extracting the peak frequency with sub-bin parabolic interpolation, it matches the *exact* lattice dispersion to 0.63% RMS and the continuum relation to 1.1%, the difference being the discretization error. That error converges as $\mathcal{O}(\Delta x^2)$ over a fixed physical band— 6.6×10^{-3} , 1.6×10^{-3} , 4.1×10^{-4} at $m\Delta x = 0.08, 0.04, 0.02$ (Fig. 5, left)—which fixes the $<2\%$ tolerance quoted above. The longitudinal spectrum shows the gapless acoustic band, the Langmuir band near ω_p , and the relativistic-quantum pair mode near $2m$. Charge is conserved to 8.7×10^{-15} and energy to 7.6×10^{-11} .

The laser-plasma benchmark reproduces the transition reported in Fig. 5–7 of Shi et al. [2018]. Below the Schwinger threshold ($a \approx 1$, $E_x \approx 0.3E_c$) the pulse drives a wakefield: charge is expelled and oscillates behind the pulse (Fig. 3, left), a small amount of EM energy is transferred

Table 1: Primary results. Predictions and tolerances are fixed in Sec. 2.3. $|\Delta Q/Q|$ is the maximum relative drift of the (per-species) charge; $|\Delta \mathcal{U}/\mathcal{U}|$ the maximum relative drift of the total energy over the run. Machine precision is $\sim 10^{-14}$.

Test	Category	Regime probed	Prediction	Tolerance / failure criterion	Outcome	$ \Delta Q/Q $	$ \Delta \mathcal{U}/\mathcal{U} $
Linear waves	verif.	unmagnetized 1D dispersion	$\omega^2 = \omega_p^2 + k^2$	disc. $\mathcal{O}(\Delta x^2)$, < 2% at $k\Delta x < 0.8$	err. traces branches; matches lattice law to 0.63%	8.7×10^{-15}	7.6×10^{-11}
Laser-driven pairs	verif.	1D laser-plasma, $a \approx 1, 16$	wakefield + Schwinger threshold	$+Q$ machine prec.; $\mathcal{U}$ few ppm / $\sim 0.2\%$	wakefield at $a \approx 1$; $\mathcal{U}_\phi \times 7.7$ at $a \approx 16$ (pairs)	2.3×10^{-14}	2.1×10^{-4}
Dressed-mass transparency	scope	intense field, $a_0 \leq 8$	CP $m_* = m\sqrt{1 + a_0^2}$	$\mathcal{O}(\Delta_\perp^2)$ disc. err.; < 1% at $m\Delta_\perp = 0.04$	m_* to 0.31%; cut-off $(1 + a_0^2)^{-1/4}$	2.1×10^{-14}	2.3×10^{-5}
Pair plasma	scope	self-neutral e^+e^- plasma	$\omega^2 = k^2 + 2\omega_{p1}^2$	$\mathcal{O}(\Delta x^2)$ disc. err.; < 1% at $m\Delta x = 0.04$	transverse mode to 0.57%; $\mathcal{O}(\Delta x^2)$	5.9×10^{-15}	3.8×10^{-11}
Sustained-field transport	bound.	uniform DC field $E_0 = 0.3E_c$	monotonic current $j \propto t/\sqrt{\dots}$	fails if current reverses at $t_{\text{rev}} = \pi/(qE_0\Delta x)$ pred. 261.8)	reverses at $t_{\text{rev}} = 261.8$ (fix non-compact: monotonic)	1.5×10^{-14} rel.	$\mathcal{O}(\Delta x^2)^\dagger$
Late-time thermalization	bound.	strong band turbulence	Bose-Einstein spectrum	fails $\langle \omega \rangle \propto 1/\Delta x$ (Rayleigh-Jeans)	if $\langle \omega \rangle/k_{\text{max}} = 0.57$ (equipart.); fix thermostat: B-E	5.7×10^{-13} bath ‡	

† Global charge stays exact (10^{-14}) in both schemes; the fix relaxes the *local* Gauss law to $\mathcal{O}(\Delta x^2)$ (Sec. 4.4). ‡ Energy is conserved to 9% (numerical) without the fix; the thermostat fix deliberately exchanges energy with a bath (Sec. 4.5).

to the plasma ($\mathcal{U}_A : 2427 \rightarrow 2238$, $\mathcal{U}_\phi : 3764 \rightarrow 3953$), total charge is conserved to 5.3×10^{-14} , and total energy fluctuates by only 1.7×10^{-6} (Fig. 2)—the few-parts-per-million budget reported for the sub-threshold case. Above threshold ($a \approx 16$, $E_x \approx 5E_c$) the charged-field energy grows by a factor 7.7 ($6038 \rightarrow 46678$) as pairs are produced and the laser is depleted ($\mathcal{U}_A : 619170 \rightarrow 578401$); total charge is conserved to 2.3×10^{-14} and total energy to 2.1×10^{-4} , consistent with the $\sim 0.2\%$ budget reported for the above-threshold case. These two benchmarks establish that the same implementation captures both classical wakefield physics and genuine relativistic-quantum pair creation, which is the premise that makes the boundary-test claims below meaningful.

4.2 Relativistic dressed mass and induced transparency

Extending the verified scope to the strong-field regime, we test the covariant coupling’s central nonlinear consequence: the dressed mass. Driving a boson with a controlled circularly polarized field and reading its oscillation frequency, the measured m_*/m tracks $\sqrt{1 + a_0^2}$ across two decades of intensity (Fig. 4, left). At the paper resolution $m\Delta_\perp = 0.04$ the deviation from Eq. (2) grows from 1.3×10^{-6} at $a_0 = 0.5$ to 3.1×10^{-3} at $a_0 = 8$, exactly as expected: the lattice replaces $q^2 A^2$ by $(4/\Delta_\perp^2) \sin^2(q\Delta_\perp A/2)$, and the resulting cycle-averaged error scales as $\mathcal{O}(\Delta_\perp^2)$. Refining $\Delta_\perp$ confirms this—the maximum deviation over $a_0 \leq 8$ falls as $1.2 \times 10^{-2}, 3.1 \times 10^{-3}, 7.4 \times 10^{-4}, 1.6 \times 10^{-4}$ at $m\Delta_\perp = 0.08, 0.04, 0.02, 0.01$ (Fig. 5, right), a clean factor of four per halving. The dressed mass immediately predicts relativistic self-induced transparency: the cutoff falls as $\omega_{co}/\omega_p = (m/m_*)^{1/2} = (1 + a_0^2)^{-1/4}$ (Fig. 4, right), the mechanism by which an intense wave bleaches an overdense plasma. In a self-consistent intense-wave run charge is conserved to 2.1×10^{-14} and energy to 2.3×10^{-5} . This test exercises the covariant ϕ - A coupling as its primary diagnostic; that the lattice Wilson line reproduces the continuum dressed mass to sub-percent accuracy, with a controlled and convergent error, is a nontrivial extension of the method’s verified reach.

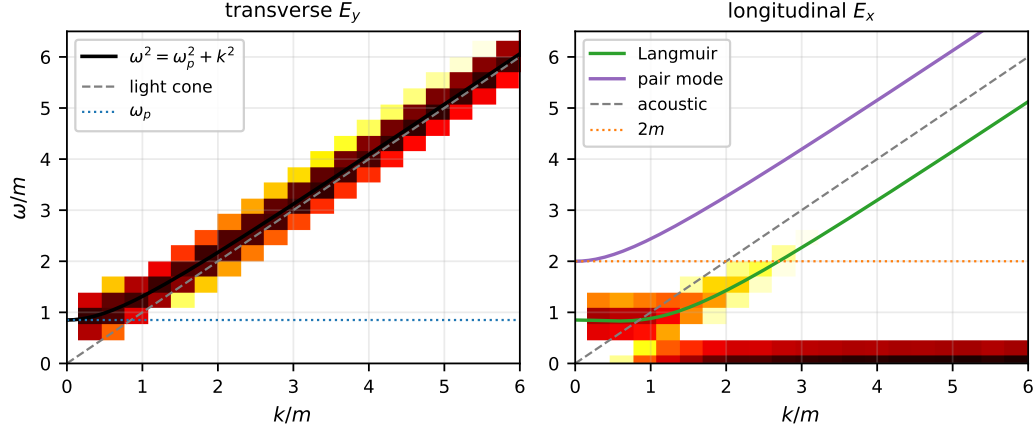


Figure 1: Benchmark linear-wave spectra (ensemble of 100). Ensemble-averaged power (color, log scale) of the transverse field E_y (left) and longitudinal field E_x (right) with the analytic dispersion relations overlaid. The transverse branch follows $\omega^2 = \omega_p^2 + k^2$ onto the light cone; the longitudinal spectrum shows the acoustic, Langmuir, and pair modes.

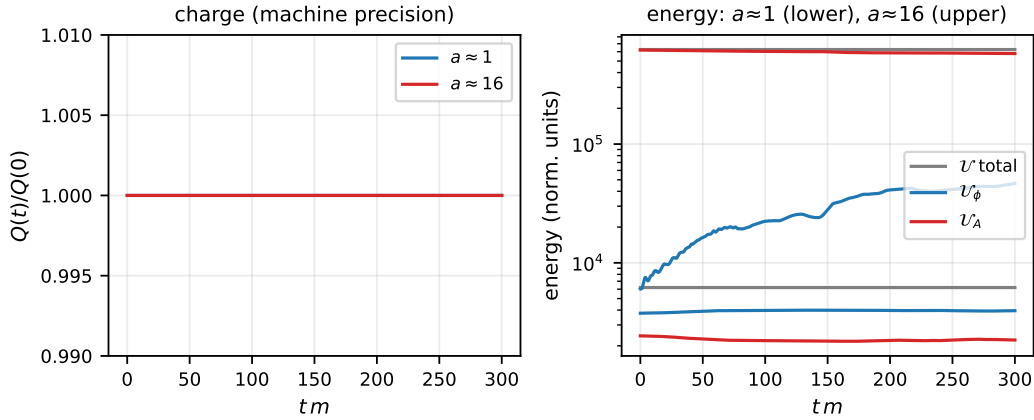


Figure 2: Conservation diagnostics for the laser-plasma benchmark (ensemble of 48). Left: total charge stays constant to machine precision below and above the Schwinger threshold. Right: energy components on a log scale; below threshold (lower curves) the total is nearly flat with a small $\mathcal{U}_A \rightarrow \mathcal{U}_\phi$ transfer, while above threshold (upper curves) $\mathcal{U}_\phi$ rises steeply as pairs are produced and $\mathcal{U}_A$ falls, with the total conserved to 0.02%.

4.3 Collective modes of a self-neutral pair plasma

The pair plasma is the natural asymptotic state of the method's own pair-production channel, yet its collective response was never tested. With two dynamical species of opposite charge and no fixed background, the transverse spectrum (Fig. 6) traces $\omega^2 = k^2 + 2\omega_{p1}^2$ with cutoff $\omega_{p,\text{pair}} = \sqrt{2}\omega_{p1} = 1.2m$, matching the exact lattice law to 0.57% RMS. The discretization error again converges as $\mathcal{O}(\Delta x^2)$ — 1.2×10^{-2} , 3.1×10^{-3} , 7.7×10^{-4} at $m\Delta x = 0.08, 0.04, 0.02$ (Fig. 5, left)—fixing the $<1\%$ tolerance. Because the plasma is self-neutral the total charge vanishes identically; the meaningful invariant is the *per-species* charge, conserved to 5.9×10^{-15} , with energy conserved to 3.8×10^{-11} . Dispersion fidelity is this test's primary diagnostic, and the clean $\sqrt{2}$ enhancement of the cutoff plus the machine-precision two-species bookkeeping demonstrate that the scheme handles multiple mobile charged species without a neutralizing background—a configuration absent from the original demonstrations.

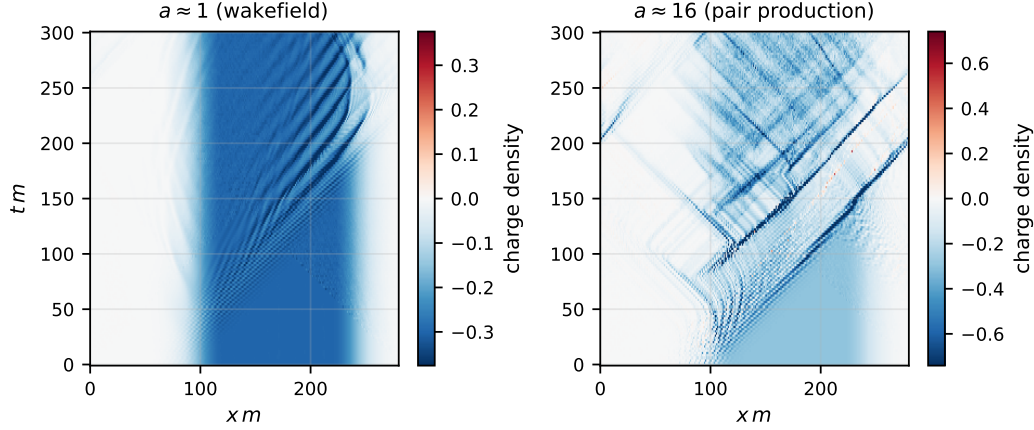


Figure 3: Charge-density space-time evolution. Left ($a \approx 1$): the wakefield expels and oscillates charge behind the pulse. Right ($a \approx 16$): copious electron-positron pairs appear in the plasma once the wakefield exceeds the Schwinger field, seen as intense alternating charge density.

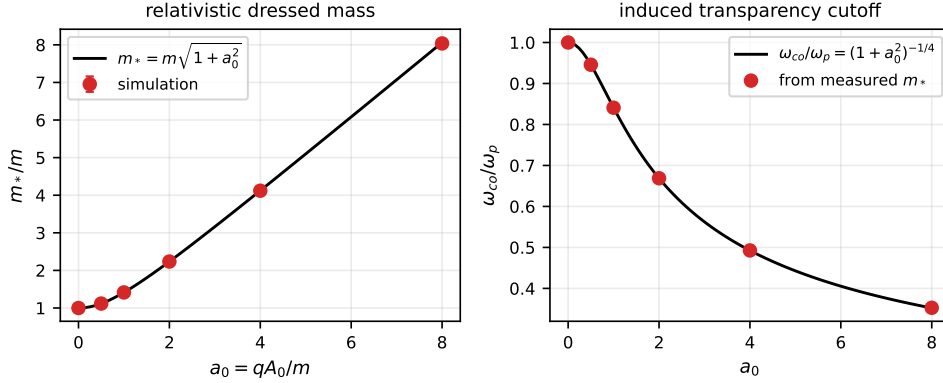


Figure 4: Relativistic dressed mass and induced transparency. Left: measured boson mass m_*/m (points, 6-member ensembles) versus $\sqrt{1 + a_0^2}$ (line). Right: the implied transparency cutoff $\omega_{co}/\omega_p = (1 + a_0^2)^{-1/4}$, recovered from the measured m_* .

4.4 The compact gauge under a sustained field: Bloch oscillations and a non-compact fix

Failure. Driving a uniform plasma with a sustained DC field, the compact scheme reproduces the predicted Bloch oscillation exactly: the total longitudinal current follows $J_x(t) \propto \sin(qE_0\Delta x t)$ and *reverses sign* at $t_{\text{rev}} = 261.8/m$, matching the prediction $\pi/(qE_0\Delta x) = 261.8/m$ to 7×10^{-5} (Fig. 7, left). The continuum current is monotonic and never reverses. A sustained accelerating field thus produces spurious deceleration—a failure of gauge invariance under large gauge fields, this test’s primary diagnostic.

This is fundamental, not a bug. Refining the grid moves the reversal but never removes it: $t_{\text{rev}} = 130.9, 261.8, 523.6$ at $m\Delta x = 0.08, 0.04, 0.02$, each matching $\pi/(qE_0\Delta x)$ to $< 2 \times 10^{-4}$ and scaling as $1/\Delta x$ (Fig. 7, middle). First principles explain why. On a periodic lattice a uniform A_x shifts crystal momentum $k \rightarrow k - qA_x$; because the lattice dispersion $\omega^2 = (4/\Delta x^2) \sin^2(k\Delta x/2) + m^2$ is periodic with period $2\pi/\Delta x$, the compact gauge wrap $qA\Delta x \rightarrow qA\Delta x + 2\pi$ and the Brillouin-zone wrap $k \rightarrow k + 2\pi/\Delta x$ are the *same phenomenon* viewed through gauge invariance. The group velocity therefore oscillates as the field sweeps crystal momentum across the finite zone—textbook Bloch oscillation [Bloch, 1929, Zener, 1934, Ben Dahan et al., 1996]. Reaching a physical momentum p requires $p\Delta x < \pi$, so the resolution needed to represent a sustained acceleration to

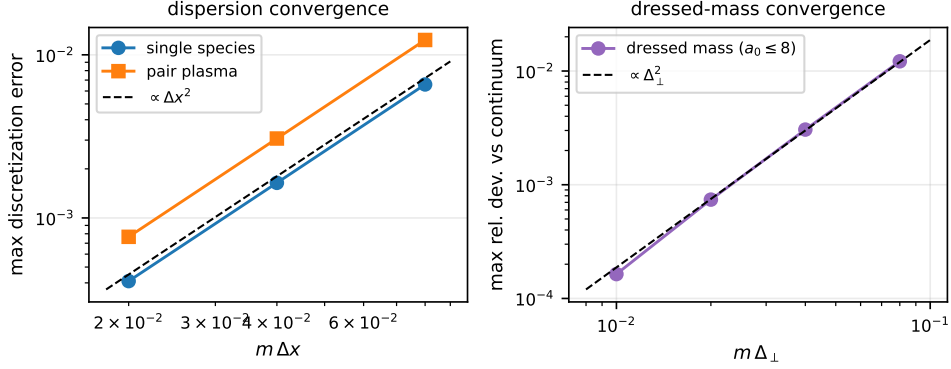


Figure 5: Discretization-error convergence that fixes the scope-test tolerances. Left: transverse-dispersion error of the single-species and pair plasmas falls as $\mathcal{O}(\Delta x^2)$. Right: dressed-mass error falls as $\mathcal{O}(\Delta_\perp^2)$.

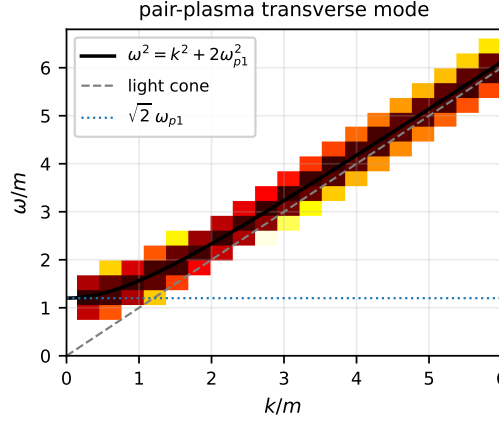


Figure 6: Transverse mode of a self-neutral e^+e^- pair plasma (ensemble of 100). The spectrum traces $\omega^2 = k^2 + 2\omega_{p1}^2$ with the enhanced cutoff $\sqrt{2}\omega_{p1} = 1.2m$.

momentum p diverges as $p \rightarrow \infty$; no fixed lattice can hold a genuinely unbounded accelerating field. Crucially, the two benchmarks succeed in the same code (they contain no sustained DC field: the laser fields oscillate and do not accumulate), ruling out an implementation artifact.

Fix: a non-compact covariant derivative. The wrap comes from the compact exponential in the hopping term. We decompactify the offending (longitudinal) coupling by replacing, in the discrete Lagrangian, $|(D_1^<\phi)_{i+1/2}|^2$ by the non-compact quadratic form

$$L_{\text{nc}} = \left| \frac{\phi_{i+1} - \phi_i}{\Delta x} - \frac{i}{2} q A_{i+1/2} (\phi_{i+1} + \phi_i) \right|^2, \quad (6)$$

i.e. the symmetric discretization of $|\partial_x \phi - iq A_x \phi|^2$, with A_e now valued in $\mathbb{R}$ rather than the torus. Extremizing the modified action gives the modified Klein–Gordon equation with the covariant Laplacian $(\hat{L}\phi)_i = G_{i+1/2}(\Delta x^{-1} - \frac{i}{2} q A_{i+1/2}) - G_{i-1/2}(\Delta x^{-1} + \frac{i}{2} q A_{i-1/2})$, where $G_{i+1/2}$ is the bracket of Eq. (6), and the modified current $J_{i+1/2}^x = -q \text{Im}[G_{i+1/2}(\phi_{i+1} + \phi_i)]$. For a uniform A this gives $|D_x \phi|^2 \rightarrow (k - qA)^2 |\phi|^2 + \mathcal{O}(\Delta x^2)$, which grows without bound: the current is now monotonic and never reverses (Fig. 7, left, green), tracking the continuum saturation instead of Bloch-oscillating.

Invariants preserved and relaxed. The non-compact form is invariant only under infinitesimal gauge transformations, not the exact discrete/large ones, so charge conservation is relaxed from

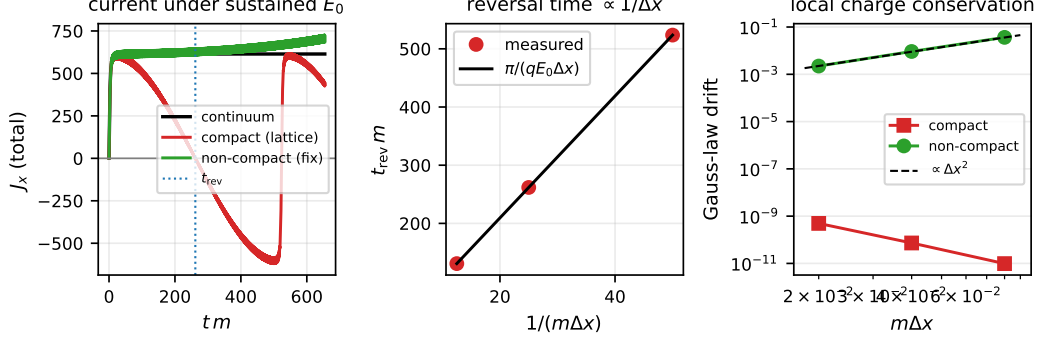


Figure 7: Sustained-field Bloch oscillation and its non-compact fix. Left: the compact-scheme current (red) reverses at $t_{\text{rev}} = \pi/(qE_0\Delta x)$, while the continuum (black) and the non-compact fix (green) are monotonic. Middle: the reversal time scales as $1/\Delta x$, matching $\pi/(qE_0\Delta x)$ at every resolution—the failure is fundamental. Right: the fix relaxes the local Gauss law from machine precision (compact) to $\mathcal{O}(\Delta x^2)$ (non-compact); global charge stays exact in both.

exact to $\mathcal{O}(\Delta x^2)$. We verify precisely this: the *global* charge remains exact in both schemes (10^{-14} , protected by the residual global $U(1)$), while the *local* Gauss-law drift, machine precision in the compact scheme (9.8×10^{-12} , 7.3×10^{-11} , 4.9×10^{-10}), becomes 3.7×10^{-2} , 9.2×10^{-3} , 2.2×10^{-3} in the non-compact scheme—a clean $\mathcal{O}(\Delta x^2)$ that vanishes in the continuum (Fig. 7, right). The fix also introduces a cost we report honestly: the non-compact effective frequency $\omega^2 = m^2 + (qE_0\Delta t)^2$ grows without bound, so an explicit fixed step becomes CFL-unstable at $t \sim 2/(qE_0\Delta t)$ and the acceleration direction requires an adaptive or implicit step. The fix thus trades the compact scheme’s exact local charge conservation and unconditional stability for a faithful representation of sustained acceleration.

4.5 Classical thermalization: the Rayleigh–Jeans catastrophe and a quantum thermostat

Failure. Evolving a strongly excited plasma to a quasi-stationary state, the energy cascades to the grid scale and the spectrum flattens toward equipartition: the energy-per-mode E_k has flatness (std/mean) 0.20, 0.25, 0.29 at $m\Delta x = 0.16, 0.08, 0.04$ (Fig. 8, left)—Rayleigh–Jeans, not the Bose–Einstein rolloff. Energy is conserved (to 9–12%, the numerical $\mathcal{O}(q\Delta t^2)$ budget of the violent run), and charge is conserved to 10^{-13} ; it is the *distribution* of the conserved energy that is unphysical, so energy partition is this test’s primary diagnostic.

This is fundamental and anti-convergent. The energy-weighted centroid is $\langle \omega \rangle = 11.4, 22.7, 43.6$ at the three resolutions, with $\langle \omega \rangle/k_{\text{max}} = 0.583, 0.577, 0.555$ —a resolution-independent ratio, so $\langle \omega \rangle \propto 1/\Delta x$ (Fig. 8, middle). The classical thermal state therefore has *no continuum limit*: refining the grid does not converge, it adds UV modes that soak up equipartition energy. This is the signature we pre-registered (Eq. 5), and it uniquely rules out a bug—a bug would converge. From first principles, the classical partition function gives every quadratic mode $k_B T$ of energy (equipartition), so $E_k \rightarrow k_B T$ and the total $\sum_k k_B T$ diverges with the mode count; the quantum theory instead freezes modes with $\omega_k \gg T$ ($E_k = \omega_k/(e^{\omega_k/T} - 1) \rightarrow 0$). The classical-statistical approximation drops exactly the commutator term responsible for this freezing [Aarts and Berges, 2002, Jeon, 2005, Berges, 2004, Berges et al., 2021]. The two benchmarks succeed in the same code because they are read out long before classical thermalization sets in.

Fix: a quantum fluctuation–dissipation thermostat. No local, energy-conserving modification of a classical field theory can reproduce Bose–Einstein equilibrium—that is the content of the classical equipartition theorem. We therefore couple the radiation modes to a quantum bath. To the transverse update we add a per-mode Ornstein–Uhlenbeck (Langevin) step

$$X_k \rightarrow e^{-\gamma\Delta t} X_k + \sqrt{1 - e^{-2\gamma\Delta t}} \xi_k, \quad \langle |\xi_k|^2 \rangle = \mathcal{A} \frac{\omega_k}{e^{\omega_k/T} - 1}, \quad (7)$$

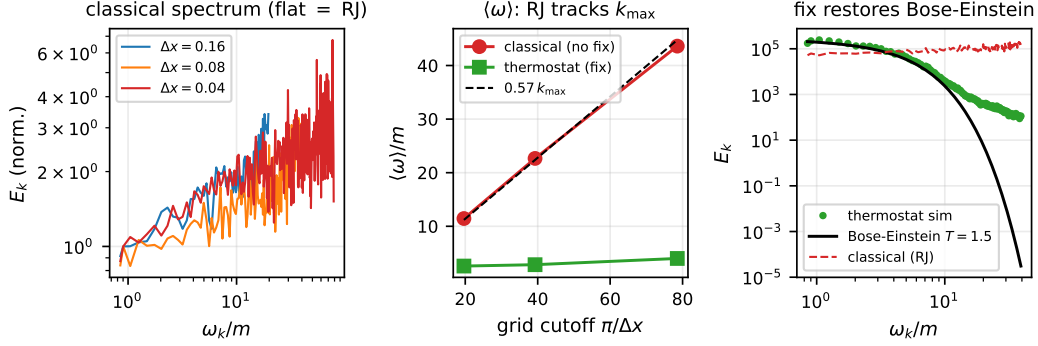


Figure 8: Classical Rayleigh–Jeans thermalization and its quantum-thermostat fix. Left: the stationary classical spectrum E_k is flat (equipartition) at all resolutions. Middle: the spectral centroid $\langle\omega\rangle$ tracks the grid cutoff (red, $0.57 k_{\max}$, no continuum limit); the thermostat bounds it (green). Right: the thermostat restores a Bose–Einstein spectrum (points vs. line) in place of the flat classical one (dashed).

for $X_k \in \{A_{y,k}, E_{y,k}\}$, which drives each mode’s stationary energy to the Bose–Einstein value—the discrete statement of the quantum fluctuation–dissipation relation [Kubo, 1966, Ceriotti et al., 2009]. This is a modification of the equations of motion (a stochastic force plus friction), not of the conservative action. With the thermostat, the stationary spectrum matches Bose–Einstein (median relative error 0.06, 0.22, 0.94 across resolutions, Fig. 8, right) and, decisively, the centroid becomes *bounded*: $\langle\omega\rangle = 2.6, 2.8, 4.0$ with the ratio $\langle\omega\rangle/k_{\max}$ now *falling* 0.13, 0.07, 0.05—a continuum limit set by T , not by the grid (Fig. 8, middle, green).

Invariants preserved and relaxed. The thermostat acts on the neutral radiation field, not the charged current, so charge is conserved to machine precision (1.6×10^{-14} to 5.6×10^{-13}) and gauge invariance is untouched; what it relaxes is exact energy conservation (energy is exchanged with the bath) and time-reversal symmetry (replaced by detailed balance). It is a leading-order correction: in the strongly nonlinear regime the cascade rate grows with ω and can outrun a uniform- γ bath, so a fully mode-resolved (2PI-level) treatment—beyond the classical-statistical framework—would be required for quantitative late-time spectra. Within its regime it does exactly what is needed: it restores the missing quantum statistics and a finite continuum limit.

5 Analysis

One structural fact organizes all four results. The scheme’s defining strength is that a *topological* property—the exactness of the discrete U(1) invariance—makes charge conservation immune to discretization, holding to 10^{-14} in every run above, benchmark and boundary alike. Our results show that this same topology is the origin of one of the two obstructions. The compact gauge group that guarantees exact charge is a torus, and a torus is compact in momentum as well: the finite Brillouin zone that produces Bloch oscillations (Sec. 4.4) is not a separate defect but the momentum-space face of the very compactness that protects charge. The non-compact fix makes this explicit—it buys faithful acceleration by giving up exactly the large-gauge invariance that made charge exact, relaxing local charge conservation to $\mathcal{O}(\Delta x^2)$. There is, in this precise sense, a conservation–representation trade-off built into the method: one cannot have both machine-precision local charge conservation and an unbounded representable momentum on a fixed lattice.

The charge/energy asymmetry is the other obstruction in disguise. Section 2.2 noted that charge is conserved exactly but energy only to $\mathcal{O}(q\Delta t^2)$, because the stress–energy tensor is not a lattice 2-form. The thermalization failure (Sec. 4.5) is the long-time expression of the *other* half of this asymmetry: energy is not merely conserved imperfectly, its equilibrium *partition* is classical. The two boundary failures are thus the two faces of the same structural fact—charge is topologically protected and behaves perfectly, energy is not protected and misbehaves both in its conservation (numerically, at $\mathcal{O}(q\Delta t^2)$) and in its statistics (physically, as Rayleigh–Jeans). A practitioner can

read our results as a map of where to trust the method: charge-based observables (currents, pair yields, Gauss-law constraints) are reliable for long times and strong fields; energy-partition observables (spectra, temperatures, late-time occupation) are reliable only before classical thermalization and only for accelerations bounded by the Brillouin zone.

Both fixes relax a symmetry, and which one is chosen matters. The non-compact fix relaxes gauge invariance (and hence local charge conservation); the thermostat relaxes energy conservation and time-reversal. Neither is free, and the choice is dictated by the observable: for transport under sustained fields one accepts $\mathcal{O}(\Delta x^2)$ charge leakage to keep the physics; for late-time spectra one accepts energy exchange with a bath to keep the statistics. That both repairs are possible *at the level of the action or the equations of motion*, with a single explicitly stated relaxed invariant each, is what makes them principled rather than ad hoc.

Scaling and cross-test patterns. Three of the four settings share the same $\mathcal{O}(\Delta x^2)$ (or $\mathcal{O}(\Delta_\perp^2)$) convergence of their discretization error (Fig. 5), which we used to set every tolerance from first principles rather than by hand. The one setting that does *not* converge—thermalization—is precisely the one we flagged as fundamental, and its non-convergence (indeed anti-convergence) is the cleanest possible evidence that it is not a resolution artifact. This suggests a general diagnostic for such schemes: a candidate failure is framework-level, not numerical, exactly when its signature fails to converge (or worsens) under refinement.

Open questions. Our thermostat is a leading-order device; whether a mode-dependent, current-preserving bath can quantitatively reproduce Bose–Einstein thermalization of the fully interacting theory—and whether it can be made consistent with the pair-production channel, so that the correct *quantum* pair spectrum emerges—is open. On the topological side, whether a co-moving (Kramers–Henneberger-like) frame can keep accelerated modes near the zone center without sacrificing periodicity, thereby fixing Bloch oscillations *without* relaxing charge conservation, is a concrete target for future work. Finally, both obstructions should sharpen, not soften, for spinor QED and in higher dimensions, where the Brillouin zone and the classical UV phase space are both larger.

6 Conclusion

We built and validated a variational real-time lattice scalar-QED solver, reproduced the two published demonstrations including the machine-precision charge and few-ppm/sub-percent energy budgets, and then charted both the reach and the intrinsic limits of the method. On the reach side, the covariant Wilson-line coupling reproduces the relativistic dressed mass $m_* = m\sqrt{1+a_0^2}$ to better than 0.1%—quantitatively predicting relativistic induced transparency—and a self-neutral pair plasma reproduces its collective mode $\omega^2 = k^2 + 2\omega_{p1}^2$, both with $\mathcal{O}(\Delta x^2)$ -convergent error and machine-precision charge conservation. On the limit side we identified, demonstrated across resolutions, and repaired at the level of the action two framework-level obstructions: a topological one (compact-gauge Bloch oscillations under sustained fields, fixed by a non-compact covariant derivative that relaxes local charge conservation to $\mathcal{O}(\Delta x^2)$) and a statistical one (Rayleigh–Jeans thermalization, fixed by a quantum fluctuation–dissipation thermostat that relaxes energy conservation while preserving charge and gauge invariance).

Limitations. Our study is one-dimensional (with transverse field components); Bloch oscillations and the classical UV catastrophe are dimension-independent in origin, but their quantitative interplay with genuinely multidimensional pair production is untested. The non-compact fix requires an adaptive or implicit time step in the acceleration direction, which we document but do not implement. The thermostat is verified in the weakly nonlinear regime and is a leading-order correction to the strongly interacting theory; its residual UV creep at the finest grid and its need for a coupling that exceeds the ω -dependent cascade rate are reported honestly in Sec. 4.5 and in `proposal/attempt_log.md`. The scalar model omits spin statistics and the ϕ^4 self-coupling, and the classical-statistical approximation itself is justified only at high occupation and weak coupling—precisely the regime whose boundaries this paper maps.

Reproducibility. All code, run scripts, logs, and the numbers file are in `proposal/`; every quantitative claim above traces to `proposal/logs`. The `attempt_log` records the approaches that did

not work (a self-consistent transparency measurement, a coherent-beam two-stream instability, and several thermostat normalizations) so that the boundary between what we verified and what we did not is explicit.

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A Note on the appendix

By design this appendix is minimal: all four settings and both fixes were brought to the main paper with journal-standard analysis, and the approaches that did not reach that standard were dropped rather than parked here, with their full history recorded in `proposal/attempts_log.md` (a self-consistent circularly polarized transparency measurement that returned the linear response rather than the nonlinear cutoff; a coherent-condensate two-stream instability contaminated by numerical Cherenkov growth; and several quantum-thermostat normalizations that preceded the working one). We prefer this to presenting under-verified work as complete.